

Regressive Financial Protection? Quantile Regression Analysis of Out-of-Pocket Health Expenditure in Peru

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Abstract

Standard analyses of catastrophic health expenditure (CHE) in developing countries rely on binary threshold indicators that mask heterogeneity across the spending distribution. Using nationally representative data from Peru's 2024 National Household Survey (ENAHO, $N = 33,691$ households), we apply conditional quantile regression at $\tau = \{0.50, 0.75, 0.90\}$ to examine whether the correlates of out-of-pocket (OOP) health expenditure share vary across the conditional distribution. We interact SIS (Seguro Integral de Salud) membership with consumption quintiles, restricting the primary analysis to the SIS-uninsured comparison given data limitations in EsSalud coverage measurement. Our results reveal a monotonic regressive protection gradient: at the median, SIS membership shows no statistically significant association with OOP share, but at upper quantiles ($\tau = 0.75, 0.90$), SIS-quintile interactions indicate larger OOP reductions for higher-quintile enrollees—a pattern consistent with regressive financial protection rather than a middle-income squeeze. Consumption quintile gradients steepen from $\tau = 0.50$ to $\tau = 0.90$, though this gradient is partly mechanical given that quintiles are constructed from total consumption inclusive of health spending. These results are provisional: bootstrap inference uses 200 replications (below the 999 standard), and the 44.1% zero-OOP mass limits the range of informative quantiles. Subject to these caveats, the findings suggest that mean-based CHE analyses understate the concentration of financial risk in the upper tail of the spending distribution.

Keywords: catastrophic health expenditure, quantile regression, health insurance, out-of-pocket spending, Peru, ENAHO

JEL codes: I13, I14, I18, C21

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1 Introduction

Catastrophic health expenditure—defined as out-of-pocket (OOP) spending exceeding a fixed share of household resources—remains a central concern for health systems in developing countries (Xu et al., 2003; Wagstaff and van Doorslaer, 2003). In Peru, the coexistence of a subsidized public insurance scheme for the poor (Seguro Integral de Salud, SIS), a contributory scheme for formal workers (EsSalud), and a large uninsured population creates a fragmented landscape in which financial protection against health shocks varies sharply across the income distribution (Petrera et al., 2013; Francke, 2013). Despite two decades of coverage expansion under the Universal Health Coverage framework established by Law 29344 (Ley Marco de Aseguramiento Universal en Salud, 2009), evidence on how effectively these schemes reduce financial risk across different segments of the spending distribution remains limited.

The standard approach in the CHE literature estimates the prevalence of households exceeding a binary threshold—typically 10%, 25%, or 40% of total consumption (Xu et al., 2003)—and examines its correlates through logistic regression at the conditional mean. This approach obscures two dimensions of heterogeneity. First, it collapses the distribution of OOP burden into a single binary indicator, discarding information about spending intensity. Second, mean-based regression assumes that covariates shift the spending distribution uniformly, a restriction that quantile regression relaxes (Koenker and Bassett, 1978; Koenker, 2005). Moreover, the large fraction of households reporting zero OOP spending—a pervasive feature of health expenditure data in developing countries—renders standard quantile regression degenerate at lower quantiles without appropriate adjustments (Powell, 1986; Manning et al., 1987).

This paper applies conditional quantile regression to Peru’s 2024 National Household Survey (Encuesta Nacional de Hogares, ENAHO), interacting insurance status with consumption quintiles to examine whether the association between insurance coverage and OOP health expenditure share varies across the conditional spending distribution. We estimate models at $\tau = \{0.50, 0.75, 0.90\}$ —quantiles at which the conditional quantile function is positive and informative given the 44.1% zero-OOP mass in our sample—and complement the analysis with a two-part model that separately identifies the extensive margin (probability of any OOP spending) and the intensive margin (spending conditional on positive expenditure).

Our analysis proceeds in four steps. First, we document the distributional features of OOP health expenditure in Peru, highlighting the zero-spending mass and the pronounced right skew of positive spending. Second, we estimate survey-weighted OLS and conditional quantile regressions with insurance–quintile interactions, demographic controls, and region fixed effects, using cluster-bootstrapped standard errors at the primary sampling unit (PSU) level. Third, we decompose the total insurance association by quintile

to identify where in the distribution insurance coverage is most strongly associated with reduced OOP burden. Fourth, we subject our findings to 25 robustness checks spanning alternative samples, CHE thresholds, consumption floor sensitivity, control sensitivity, and placebo tests.

Three provisional findings emerge, subject to the data and inference limitations detailed below. First, the base SIS coefficient is statistically indistinguishable from zero across all quantiles when evaluated at the lowest consumption quintile (Q1), consistent with the interpretation that SIS enrollees in Q1 either access free care or forgo care altogether. Second, the SIS–quintile interaction terms are negative and increasingly significant at higher quantiles and higher quintiles: at $\tau = 0.90$, SIS membership in Q5 is associated with a reduction of 1.94 percentage points in OOP share ($p < 0.01$ with 200-replication bootstrap; significance is provisional pending 999-replication confirmation). This monotonic gradient—where the insurance association *increases* with income conditional on enrollment—is more accurately characterized as a regressive protection pattern than as a middle-income squeeze. Third, consumption quintile gradients steepen monotonically across quantiles, with the Q5 coefficient rising from near zero at $\tau = 0.10$ to 5.16 percentage points at $\tau = 0.90$. However, this gradient is partly mechanical: consumption quintiles are constructed from GASHOG2D, which includes health expenditure, creating a positive correlation between quintile rank and OOP share. Reconstruction of quintiles from non-health consumption is required to isolate the behavioral gradient from the mechanical component.

We restrict the primary analysis to the SIS–uninsured comparison. Our EsSalud sample ($N = 648$ household heads, approximately 1.9% of the sample) is implausibly small relative to national EsSalud coverage rates of approximately 30%, likely reflecting a variable construction error in the household-head-based insurance assignment that discards non-head EsSalud members. This misclassification contaminates the uninsured reference group with households that have EsSalud coverage through non-head members. EsSalud results are reported in supplementary tables but are excluded from substantive interpretation.

We interpret the monotonic SIS–quintile gradient cautiously. SIS is means-tested through the SISFOH (Sistema de Focalización de Hogares), and its presence in upper quintiles reflects well-documented targeting errors and leakage (Cortez, 2008; Francke, 2013). The larger SIS–quintile interactions at Q4–Q5 may reflect a composition effect—non-poor SIS enrollees who access care more effectively than their poor counterparts—rather than evidence that SIS provides differential financial protection by design. Our descriptive framework cannot distinguish between these mechanisms.

These findings contribute to three literatures. First, we add to the growing body of work applying quantile regression to health expenditure analysis (Jones et al., 2013; Machado and Santos Silva, 2019), demonstrating that the conditional correlates of OOP

burden are quantile-dependent in a middle-income country context. Second, we contribute to the literature on Peru’s health financing system ([Espinoza-Pajuelo et al., 2024](#); [Petrera et al., 2013](#)), providing the first distributional analysis of CHE determinants using the most recent ENAHO wave. Third, we inform the policy debate on universal coverage design by showing that average CHE rates mask substantial heterogeneity in financial risk across both the income and spending distributions.

We emphasize throughout that our estimates represent conditional associations, not causal effects. Insurance enrollment in Peru is non-random: SIS targets the poor through means testing, while EsSalud covers formal-sector workers ([Wagstaff, 2010](#); [Acharya et al., 2013](#)). We note further that the 2024 ENAHO is collected during the post-pandemic recovery period; Peru experienced among the world’s highest COVID-19 excess mortality rates, with substantial disruption to health-seeking behavior. Health expenditure patterns in 2024 may reflect post-pandemic adjustment rather than steady-state behavior, a caveat that limits the generalizability of our findings.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data and variable construction. Section 4 presents the empirical strategy. Section 5 reports the main results. Section 6 discusses robustness checks. Section 7 concludes.

2 Literature Review

2.1 Catastrophic Health Expenditure in Developing Countries

The conceptual framework for catastrophic health expenditure originates with [Wagstaff and van Doorslaer \(2003\)](#) and [Xu et al. \(2003\)](#), who define CHE as OOP spending exceeding a fixed share of household resources—total consumption or capacity-to-pay (consumption net of subsistence food expenditure). This binary threshold approach has been applied extensively across low- and middle-income countries, generating a large body of evidence on CHE prevalence and its correlates ([Wagstaff, 2008](#); [World Health Organization and World Bank, 2020](#)). Common findings include that CHE disproportionately affects the uninsured, the elderly, households with chronic illness, and those in lower income quintiles.

The binary threshold approach has well-known limitations. [Flores et al. \(2008\)](#) demonstrate that CHE prevalence is sensitive to the choice of threshold and denominator, while [Moreno-Serra et al. \(2011\)](#) argue that mean-based regression on the binary indicator discards information about spending intensity. Households spending 11% and 90% of consumption on health are treated identically if the threshold is 10%, yet their financial distress differs qualitatively.

2.2 Quantile Regression in Health Expenditure Analysis

Quantile regression, introduced by [Koenker and Bassett \(1978\)](#) and formalized in [Koenker \(2005\)](#), estimates the conditional quantile function, allowing covariate effects to vary across the distribution. In health expenditure analysis, this is particularly valuable because the OOP distribution is typically zero-inflated, right-skewed, and heteroskedastic.

[Jones et al. \(2013\)](#) survey quantile regression methods for health economics applications, emphasizing the distinction between conditional and unconditional quantile effects. [Machado and Santos Silva \(2019\)](#) develop a quantile regression framework for count data with applications to health care utilization. In the developing-country context, [Brendenkamp et al. \(2011\)](#) use quantile regression to analyze OOP spending in the Western Balkans, documenting that insurance effects are concentrated in the upper tail.

A critical methodological challenge arises when the outcome has a mass point at zero. Standard Koenker–Bassett quantile regression is degenerate at quantiles below the zero-mass fraction, producing uninformative estimates ([Machado and Santos Silva, 2005](#)). Two approaches address this. First, censored quantile regression ([Powell, 1986](#); [Chernozhukov and Hong, 2002](#)) treats zero as a censoring point, producing consistent estimates of the latent quantile function. Second, the two-part model ([Manning et al., 1987](#); [Mullahy, 1998](#)) separates the extensive margin (participation) from the intensive margin (spending conditional on participation), allowing different covariates to drive each decision. We adopt the two-part framework as a complement to our conditional quantile regression, restricting interpretation of CQR results to quantiles above the zero-mass threshold.

The recentered influence function (RIF) regression of [Firpo et al. \(2009\)](#) estimates unconditional quantile effects—how a covariate shifts the population-level quantile. While conditional quantile regression answers “at what point in the conditional distribution does a covariate matter?”, RIF regression answers the population-level question implied by many policy discussions. We note this distinction but focus on conditional quantile regression, as the insurance–quintile interaction effects we estimate are inherently conditional objects. A RIF extension would provide the population-level complement and is a natural direction for future work.

2.3 Health Financing in Peru

Peru’s health system is characterized by segmentation. SIS, created in 2002, provides subsidized coverage to households in poverty and extreme poverty, covering primary care and selected hospital services at public facilities ([Petrera et al., 2013](#)). The legal framework for universal coverage was established by Law 29344 (Ley Marco de Aseguramiento Universal en Salud, 2009), which created the Plan Esencial de Aseguramiento en Salud (PEAS) as the minimum benefit package. EsSalud, the social security scheme, covers formal-sector workers and their dependents, providing more comprehensive benefits. Pri-

vate insurance (EPS) covers a small fraction of the population, primarily in Lima.

[Espinoza-Pajuelo et al. \(2024\)](#) analyze informal payments in Peruvian health facilities using the 2018 ENAHO, finding that SIS beneficiaries incur undisclosed costs for nominally free services. [Petrera et al. \(2013\)](#) document persistent OOP spending among SIS enrollees, particularly for medications and diagnostic services not covered by PEAS. In the Latin American context, [Knaul et al. \(2006\)](#) study Mexico’s Seguro Popular—a subsidized insurance scheme targeting the poor, analogous to SIS—finding that coverage expansion reduced CHE prevalence but with heterogeneous effects across the income distribution.

A critical institutional feature is SIS targeting imprecision. The SISFOH system uses proxy means testing to determine eligibility, but substantial leakage—enrollment of non-poor households—has been documented ([Cortez, 2008](#); [Francke, 2013](#)). This leakage complicates the interpretation of SIS effects at upper consumption quintiles: positive SIS enrollment at Q4–Q5 may reflect targeting errors rather than legitimate enrollment, and any measured financial protection at these quintiles conflates the insurance effect with the composition of the (mis)targeted population.

2.4 Identification Challenges with Observational Insurance Data

Estimating the effect of health insurance on OOP spending faces fundamental identification challenges. Insurance enrollment is non-random: means-tested programs like SIS select on poverty status, while contributory schemes like EsSalud select on formal employment. [Wagstaff \(2010\)](#) formalizes these challenges for observational studies and surveys quasi-experimental approaches (RDD at eligibility thresholds, enrollment mandates) that can identify causal effects under specific assumptions. [Acharya et al. \(2013\)](#) provide a systematic review of health insurance impact evaluations in low- and middle-income countries, finding that observational studies consistently overestimate protective effects due to uncontrolled selection.

Our paper is explicitly descriptive. We do not claim to identify causal effects of insurance on OOP spending. Instead, we document conditional distributional patterns that characterize where financial protection gaps are concentrated—information that binary CHE indicators and mean-based regressions cannot deliver, but that stops short of causal attribution.

3 Data

3.1 Source

We use Peru’s 2024 National Household Survey (Encuesta Nacional de Hogares, ENAHO), a nationally representative repeated cross-sectional survey conducted by the National Institute of Statistics and Informatics (INEI). ENAHO employs a stratified, multistage probability sampling design with primary sampling units (PSUs, identified by the variable `CONGLOME`) and strata defined by geographic domain and urban/rural classification. We use the survey expansion factor `FACTOR07` from the Sumaria module throughout all descriptive statistics and regression models to produce population-representative estimates.

We note that the 2024 ENAHO is collected during the post-pandemic recovery period. Peru experienced among the highest COVID-19 excess mortality rates globally, with substantial disruption to health care access and health-seeking behavior. Health expenditure patterns in 2024 may reflect post-pandemic adjustment rather than steady-state behavior, a caveat that limits the temporal generalizability of our findings.

Our analysis draws on two ENAHO modules. Module 400 (Health) provides individual-level information on health care utilization, insurance status, and OOP health expenditure for 110,451 individuals. The Sumaria module provides household-level total consumption (`GASHOG2D`), which serves as the denominator for our OOP share measure. We merge these modules at the household level using the triple key `CONGLOME`×`VIVIENDA`×`HOGAR`.

3.2 Variable Construction

Outcome variable. Our primary outcome is the OOP health expenditure share, defined as:

$$\text{OOP_share}_h = \frac{\text{Health_expenditure}_h}{\max(\text{Total_consumption}_h, \text{Floor})} \quad (1)$$

where `Health_expenditure` is the sum of consultation fees (`P41601`), medicine costs (`P41602`), and other health costs (`P41603`), aggregated from individual to household level and annualized by multiplying by 13 (52 weeks ÷ 4-week recall period). `Total_consumption` is `GASHOG2D` from Sumaria, which is already annualized. The floor is set at the 5th percentile of `GASHOG2D` among households with positive consumption to prevent extreme shares from near-zero denominators. We cap `OOP_share` at $[0, 1]$ after applying the floor.

The annualization factor of 13 assumes that 4-week recall expenditure is representative of annual spending. This assumption amplifies measurement error for households experiencing acute episodes during the recall period: a single catastrophic visit is multiplied 13-fold, potentially overstating annual OOP. We report sensitivity to an alternative

$\times 12$ annualization in Section 6 and note that this amplification is the dominant source of measurement error in the outcome variable. As an additional consideration, the P41603 category (“other health costs”) captures heterogeneous items including hospitalization, transport, dental, and optical expenses, whose high within-category variance may disproportionately drive upper-quantile estimates.

Catastrophic health expenditure. We construct binary CHE indicators at three thresholds: $CHE_{10} = \mathbf{1}[\text{OOP_share} > 0.10]$, $CHE_{25} = \mathbf{1}[\text{OOP_share} > 0.25]$, and $CHE_{40} = \mathbf{1}[\text{OOP_share} > 0.40]$.

Insurance status. We construct mutually exclusive insurance categories from Module 400: (i) SIS (P4191= 1 and P4192 $\neq$ 1), (ii) EsSalud (P4192= 1, with or without SIS; EsSalud takes precedence in overlaps), and (iii) Uninsured (neither SIS nor EsSalud). The household-level insurance variable is assigned based on the household head’s coverage.

We acknowledge an important data limitation: our EsSalud sample comprises only $N = 648$ household heads (1.9% of the sample), which is substantially below the expected 30–35% national EsSalud coverage rate. This discrepancy likely reflects a variable construction issue in the P4192 coding within Module 400—possibly related to how multiple insurance affiliations are recorded at the individual level, or to the fact that Module 400 captures a different population than the insurance-specific modules used in official coverage statistics. We were unable to resolve this discrepancy within the current analysis. Consequently, **EsSalud results throughout this paper should be interpreted with extreme caution.** The small cell sizes ($N \approx 8\text{--}130$ per quintile) render EsSalud–quintile interaction estimates imprecise and potentially unreliable. We retain EsSalud in the analysis for completeness but emphasize that the primary substantive conclusions rest on the SIS–quintile comparison.

An additional limitation of the head-based insurance assignment is that it excludes cases where the head is uninsured but dependents have SIS coverage, or vice versa. A household-level measure based on any-member coverage may better characterize financial protection. We report sensitivity to alternative aggregation rules in Section 6.

Consumption quintiles. Quintiles are constructed from survey-weighted GASHOG2D using FACTOR07-weighted quantile breaks. We note a mechanical circularity: GASHOG2D is a comprehensive consumption aggregate that, by construction, includes health expenditures (Module 400 flows into Sumaria). Households with high OOP spending are thus assigned to higher quintiles partly because of their health spending, inducing a positive correlation between OOP share and quintile rank that inflates the quintile gradient. The standard correction—constructing quintiles from consumption net of health expenditure—would require subtracting annualized health spending from GASHOG2D be-

fore computing quintile breaks. This circularity is a fundamental limitation of the current implementation: the quintile–OOP gradient reported in our results is partly mechanical rather than behavioral. Net-of-health quintiles—constructed from GASHOG2D – annualized health expenditure—are required as the primary specification to isolate the behavioral gradient. We were unable to implement this correction in the current version and flag this as a priority for reanalysis. All quintile-related results, including the insurance–quintile interactions that form the core of our analysis, should be interpreted as upper bounds on the true behavioral association.

Controls. Household head characteristics include age (P208A), sex (P207), and years of education (P301A). Household-level variables include household size, number of children under 5, number of elderly members (65+), rural residence (AREA), and presence of any chronic illness among household members. We include region fixed effects based on INEI’s 24 administrative departments (derived from the first two digits of UBIGE0).

3.3 Sample and Descriptive Statistics

After merging modules and aggregating to household level, our analysis sample comprises 33,691 households. Table 1 reports survey-weighted summary statistics. The mean OOP share is 2.7% (SD = 9.2%), with a median of 0.3%, reflecting the pronounced right skew of health spending. A substantial 41.1% of households report zero OOP health expenditure—a feature with critical implications for quantile regression, as discussed in Section 4. CHE prevalence is 5.1% at the 10% threshold, 1.3% at 25%, and 0.8% at 40%.

Table 1: Summary Statistics (Survey-Weighted)

Variable	N	Mean	SD	Min	Median	Max
oop_share	33,691	0.027	0.092	0.000	0.003	1.000
health_expend	33,691	11415.231	136853.281	0.000	65.000	3910306.400
total_consumption	33,691	31285.282	22758.677	825.153	23945.350	308555.781
zero_oop	33,691	0.411	0.492	0.000	0.000	1.000
che_10pct	33,691	0.051	0.220	0.000	0.000	1.000
che_25pct	33,691	0.013	0.113	0.000	0.000	1.000
che_40pct	33,691	0.008	0.089	0.000	0.000	1.000
age_head	33,691	53.430	15.586	15.000	53.000	98.000
female_head	33,691	0.373	0.484	0.000	0.000	1.000
educ_years	33,691	9.885	5.514	0.000	11.000	20.000
hh_size	33,691	3.344	1.760	1.000	3.000	17.000
children_u5	33,691	0.193	0.448	0.000	0.000	5.000
elderly_65	33,691	0.409	0.672	0.000	0.000	5.000
rural	33,691	0.205	0.403	0.000	0.000	1.000
chronic_any	33,691	0.163	0.369	0.000	0.000	1.000
hospitalized	33,691	0.000	0.000	0.000	0.000	0.000

Notes: ENAHO 2024. Survey-weighted (FACTOR07). Unit: household.

Table 2 presents descriptive statistics by insurance group. SIS enrollees ($N = 11,189$, 33.2% of the sample) have a mean OOP share of 3.0% and a zero-OOP rate of 37.3%. EsSalud members ($N = 648$, 1.9%) show a slightly higher mean OOP share (3.4%) and zero-OOP rate (40.9%). Uninsured households ($N = 1,165$, 3.5%) have the lowest mean OOP share (2.5%) but the highest zero-OOP rate (48.7%), consistent with care avoidance among the uninsured. SIS households are more rural (5.9% vs. 1.0% for EsSalud), have more children under 5, and lower education levels, reflecting the means-tested targeting of SIS toward poorer households.

Table 2: Descriptive Statistics by Insurance Group (Survey-Weighted)

Variable	SIS		EsSalud		Uninsured	
	Mean	SD	Mean	SD	Mean	SD
oop_share	0.030	0.104	0.034	0.136	0.025	0.074
zero_oop	0.373	0.484	0.409	0.492	0.487	0.500
health_expend	16440.741	172406.483	30701.373	235556.493	3599.961	61523.844
age_head	55.569	15.172	54.861	14.727	52.019	15.177
female_head	0.380	0.485	0.379	0.485	0.329	0.470
educ_years	11.789	5.638	13.068	7.120	9.867	5.230
hh_size	3.555	1.742	3.601	1.637	2.039	1.317
children_u5	0.171	0.425	0.136	0.368	0.036	0.202
elderly_65	0.506	0.738	0.494	0.732	0.260	0.515
rural	0.059	0.236	0.010	0.099	0.161	0.367
chronic_any	0.172	0.378	0.153	0.360	0.084	0.277
hospitalized	0.000	0.000	0.000	0.000	0.000	0.000
N	11,189		648		1,165	

Notes: ENAHO 2024. Survey-weighted means (FACTOR07).

3.4 Zero-OOP Mass and Implications for Quantile Regression

The 44.1% zero-OOP rate has direct methodological implications. At quantiles below the zero-mass fraction, the conditional quantile of OOP share is identically zero for most covariate configurations, rendering standard Koenker–Bassett quantile regression coefficients uninformative (Machado and Santos Silva, 2005). At $\tau = 0.10$ and $\tau = 0.25$, estimated coefficients reflect movements between zero and near-zero values rather than meaningful variation in spending intensity. At $\tau = 0.50$, the conditional quantile is positive for most but not all covariate cells—particularly low-income, low-utilization subgroups may still have degenerate medians.

We address this zero-mass problem through two complementary strategies. First, we restrict substantive interpretation of CQR results to $\tau \geq 0.50$, where conditional quantiles are positive for most household types, and treat $\tau = 0.10$ and $\tau = 0.25$ estimates as uninformative lower-bound specifications reported for completeness. Second, we com-

plement the CQR analysis with a two-part model that separately identifies the extensive margin (probability of any OOP spending via probit) and the intensive margin (QR on the positive-OOP subsample), following [Manning et al. \(1987\)](#). We note that censored quantile regression ([Powell, 1986](#); [Chernozhukov and Hong, 2002](#)) offers an alternative approach that treats zero as a censoring point and produces consistent estimates at all quantiles; this remains a direction for future work.

4 Empirical Strategy

4.1 OLS Benchmark

We begin with a survey-weighted OLS regression as a benchmark:

$$\text{OOP_share}_h = \alpha + \beta \mathbf{Ins}_h + \gamma \mathbf{Q}_h + \delta (\mathbf{Ins}_h \times \mathbf{Q}_h) + \theta \mathbf{X}_h + \phi \mathbf{R}_h + \varepsilon_h \quad (2)$$

where $\mathbf{Ins}_h$ is a vector of insurance dummies (SIS, EsSalud; uninsured omitted), $\mathbf{Q}_h$ is a vector of consumption quintile dummies (Q2–Q5; Q1 omitted), $\mathbf{Ins}_h \times \mathbf{Q}_h$ is the full set of insurance–quintile interactions, $\mathbf{X}_h$ is a vector of household-level controls (age, sex, and education of household head, household size, children under 5, elderly 65+, rural, chronic illness), and $\mathbf{R}_h$ is a vector of 24 department fixed effects. Standard errors are clustered at the PSU level (**CONGLOME**).

We note that linear OLS on a fractional bounded outcome $\in [0, 1]$ can produce fitted values outside the unit interval and the linear approximation is inappropriate at extremes ([Papke and Wooldridge, 1996](#)). We report a fractional logit specification as a robustness check in Section 6. The OLS benchmark is retained for interpretive transparency and comparability with the quantile regression coefficients.

The insurance–quintile interaction is the primary specification, not an alternative. Insurance categories and quintile dummies are jointly correlated by design: SIS is means-tested toward the poor, and EsSalud covers formal workers in higher quintiles. Including both without interactions partially absorbs each other’s variation; the interaction allows the insurance association to vary freely across the income distribution.

4.2 Conditional Quantile Regression

Our main specification replaces the OLS conditional mean with the conditional τ -quantile:

$$Q_\tau(\text{OOP_share}_h \mid \mathbf{Z}_h) = \alpha(\tau) + \beta(\tau) \mathbf{Ins}_h + \gamma(\tau) \mathbf{Q}_h + \delta(\tau) (\mathbf{Ins}_h \times \mathbf{Q}_h) + \theta(\tau) \mathbf{X}_h + \phi(\tau) \mathbf{R}_h \quad (3)$$

where $Q_\tau(\cdot)$ denotes the τ -th conditional quantile function ([Koenker and Bassett, 1978](#)). We estimate this model at $\tau = \{0.10, 0.25, 0.50, 0.75, 0.90\}$, but restrict substantive in-

terpretation to $\tau \geq 0.50$ given the 44.1% zero-OOP mass (Section 3). At $\tau = 0.10$ and $\tau = 0.25$, the conditional quantile is at or near zero for most covariate configurations, and coefficients are uninformative for the distributional question of interest.

The coefficient $\beta_{\text{SIS}}(\tau)$ represents the change in the conditional τ -quantile of OOP share associated with SIS membership (relative to uninsured) at Q1, holding other covariates constant. The total SIS association at quintile q is $\beta_{\text{SIS}}(\tau) + \delta_{\text{SIS} \times q}(\tau)$.

We emphasize two interpretive caveats. First, quantile regression coefficients represent changes in the conditional τ -quantile, not average marginal effects. A household at the 10th conditional quantile is not necessarily a “low-spending household” in the unconditional sense; it is a household whose spending is low relative to what its covariates predict. Second, at $\tau = 0.10$ in our setting, negative coefficients for SIS may reflect that SIS beneficiaries are more likely to have zero spending (care avoidance or access to free care), not that SIS protects against high spending. The two-part model below disentangles these channels.

Survey weights (FACTOR07) are applied as frequency weights rescaled to preserve the original sample size while reflecting the population distribution. We acknowledge that this approach does not fully propagate the stratified multistage survey design through the quantile regression estimator. Properly design-weighted quantile regression would require a more complex implementation (Koenker, 2005). As a partial check, we compare weighted and unweighted estimates in Section 6; large discrepancies would signal that the weighting scheme materially affects conclusions.

4.3 Two-Part Model

Given the 44.1% zero-OOP mass, we complement the CQR analysis with a two-part model (Manning et al., 1987):

Part 1 (Extensive margin): Survey-weighted probit for $\Pr(\text{OOP} > 0)$ on the same regressors as Equation 2. We report average marginal effects (AME), not log-odds.

Part 2 (Intensive margin): Conditional quantile regression at $\tau = \{0.25, 0.50, 0.75, 0.90\}$ on the positive-OOP subsample ($N = 18,843$). This subsample is selected (households that seek care and incur costs), introducing a Heckman-type selection concern. We note this limitation but observe that the two-part model’s value lies in decomposing the extensive and intensive margins, not in correcting for selection per se.

This framework addresses the zero-mass problem more directly than standard CQR. Part 1 identifies whether insurance status affects the probability of incurring any OOP spending—the participation margin. Part 2 characterizes how insurance is associated with the intensity of spending among those who do spend. The decomposition is informative regardless of selection concerns because it separates two qualitatively different margins that standard CQR conflates at lower quantiles.

We note that censored quantile regression (Powell, 1986; Chernozhukov and Hong, 2002) offers a theoretically preferable alternative that treats zero as a censoring point and produces consistent estimates at all quantiles without sample selection. Implementation of the Powell (1986) or Chernozhukov–Hong (2002) three-step estimator is computationally intensive with our sample size and interaction structure, and we leave this extension for future work.

4.4 Inference

Standard errors are computed via cluster bootstrap at the PSU level (CONGLOME) with 200 replications. The cluster bootstrap resamples entire PSUs, preserving the within-cluster correlation structure.

We acknowledge that 200 bootstrap replications is below the recommended 999 for publishable inference (Cameron et al., 2008). At $\tau = 0.90$, where the effective estimation sample is the top decile of OOP share, cluster-level sample sizes are thin and bootstrap standard errors are noisier. **Significance conclusions at upper-tail quantiles ($\tau = 0.75, 0.90$) should be interpreted with caution pending replication with 999+ replications.** We report Koenker–Bassett sandwich standard errors as a comparison in Section 6 and note that the wild cluster bootstrap (Roodman et al., 2019) would provide a more robust alternative when within-cluster sample sizes at extreme quantiles are small.

4.5 Multiple Testing

Our analysis involves a large family of hypothesis tests: 5 quantiles $\times$ (2 insurance dummies + 8 interaction terms + controls), plus 25 robustness regressions. Without adjustment, some results will appear significant by chance under conventional $\alpha = 0.05$ thresholds. We do not apply a formal familywise correction (Romano–Wolf or Benjamini–Hochberg) in the current version, but flag this as a limitation: the selective highlighting of significant insurance–quintile interactions at specific quantiles without multiple testing adjustment risks false discovery inflation. We report joint F-tests for the full set of interactions at each quantile as a partial mitigation.

4.6 Identification and Limitations

Our estimates are descriptive, not causal. Three sources of endogeneity warrant explicit discussion:

1. *Non-random insurance assignment.* SIS enrollment is means-tested through SIS-FOH: eligible households self-select based on poverty status, health needs, and administrative access. EsSalud membership is tied to formal employment. Insurance

status is correlated with unobserved determinants of health spending (Wagstaff, 2010; Acharya et al., 2013). We do not claim causal identification.

2. *Quintile endogeneity.* Consumption quintiles are constructed from GASHOG2D, which includes health spending. This creates a mechanical positive correlation between quintile rank and OOP share that inflates the quintile gradient throughout our results (Section 3). The interaction terms are similarly affected.
3. *SIS targeting leakage.* SIS enrollment in upper quintiles (Q4–Q5) reflects well-documented SISFOH targeting errors (Cortez, 2008; Francke, 2013). The SIS–quintile interactions at Q4–Q5 conflate insurance protection with the composition of the (mis)targeted population. We cannot decompose legitimate enrollment from leakage in our data.

Despite these limitations, the distributional analysis provides descriptive evidence on where financial protection gaps are concentrated—information that binary CHE indicators and mean-based regressions cannot deliver.

5 Results

5.1 OLS Benchmark

Column 1 of Table 3 reports the survey-weighted OLS estimates. The SIS coefficient is small and statistically insignificant ($\hat{\beta} = 0.0032$, $SE = 0.0028$), indicating that, on average, SIS membership is not associated with a meaningfully different OOP share relative to being uninsured, conditional on quintile, demographics, and region. The EsSalud coefficient is large (0.2359) but imprecise ($SE = 0.2039$), reflecting the small EsSalud cell size ($N = 648$) discussed in Section 3; we caution against interpreting this estimate.

Consumption quintile dummies display a monotonically increasing pattern: relative to Q1, households in Q2 through Q5 have progressively higher OOP shares, with the Q5 coefficient at 2.23 percentage points ($p < 0.01$). This gradient is expected but should be interpreted with the caveat that quintile assignment includes health spending in the denominator (Section 3), which mechanically inflates the quintile–OOP share correlation.

The SIS–quintile interactions are uniformly negative but individually insignificant in OLS, with the SIS×Q5 interaction at -0.0102 ($SE = 0.0065$). The OLS R^2 is 0.0201, reflecting the high residual variance characteristic of health expenditure regressions.

5.2 Quantile Regression: Main Results

Columns 2–6 of Table 3 report the conditional quantile regression estimates at $\tau = \{0.10, 0.25, 0.50, 0.75, 0.90\}$.

Table 3: Quantile Regression: Insurance Associations with OOP Share

	OLS	$\tau = 0.1$	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.9$
SIS	0.0032 (0.0028)	0.0000 (0.0000)	0.0000 (0.0000)	-0.0000 (0.0001)	0.0013 (0.0010)	0.0047 (0.0051)
EsSalud	0.2359 (0.2039)	0.0000 (0.0000)	0.0000 (0.0003)	0.0005 (0.0006)	0.0002 (0.3127)	0.9183** (0.4551)
Q2	0.0073*** (0.0017)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0006*** (0.0001)	0.0052*** (0.0007)	0.0144*** (0.0022)
Q3	0.0118*** (0.0021)	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0022*** (0.0003)	0.0085*** (0.0008)	0.0263*** (0.0026)
Q4	0.0155*** (0.0032)	0.0000*** (0.0000)	0.0000 (0.0000)	0.0046*** (0.0004)	0.0128*** (0.0011)	0.0309*** (0.0037)
Q5	0.0223*** (0.0054)	0.0000*** (0.0000)	0.0010*** (0.0003)	0.0057*** (0.0006)	0.0188*** (0.0020)	0.0516*** (0.0048)
SIS $\times$ Q2	-0.0057 (0.0049)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0003 (0.0002)	-0.0032** (0.0016)	-0.0069 (0.0062)
SIS $\times$ Q3	-0.0057 (0.0043)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0010*** (0.0003)	-0.0036** (0.0015)	-0.0147** (0.0059)
SIS $\times$ Q4	-0.0075 (0.0050)	-0.0000*** (0.0000)	-0.0000 (0.0000)	-0.0022*** (0.0005)	-0.0065*** (0.0015)	-0.0107* (0.0063)
SIS $\times$ Q5	-0.0102 (0.0065)	-0.0000*** (0.0000)	-0.0010*** (0.0003)	-0.0017** (0.0007)	-0.0057** (0.0026)	-0.0194*** (0.0073)
EsSalud $\times$ Q2	-0.2480 (0.2041)	-0.0000 (0.0000)	-0.0000 (0.0003)	-0.0008 (0.0014)	-0.0076 (0.3118)	-0.8944* (0.4580)
EsSalud $\times$ Q3	-0.2606 (0.2039)	-0.0000*** (0.0000)	-0.0000 (0.0003)	-0.0036*** (0.0014)	-0.0039 (0.3126)	-0.9364** (0.4557)
EsSalud $\times$ Q4	-0.2547 (0.2040)	-0.0000*** (0.0000)	-0.0000 (0.0003)	-0.0046*** (0.0007)	-0.0128 (0.3125)	-0.9299** (0.4544)
EsSalud $\times$ Q5	-0.2390 (0.2041)	-0.0000*** (0.0000)	-0.0010*** (0.0004)	-0.0047*** (0.0010)	-0.0081 (0.3127)	-0.9467** (0.4546)
N	33,691	33,691	33,691	33,691	33,691	33,691
R ² /Pseudo-R ²	0.0201	-0.0000	0.0021	0.0252	0.0351	0.0426

Notes: Coefficients represent changes in the conditional τ -quantile of OOP health expenditure share, not average marginal effects. Causal interpretation is not warranted given non-random insurance assignment. OLS is survey-weighted (FACTOR07) with cluster-robust SEs. QR uses cluster-bootstrapped SEs (200 reps, PSU-level). Q1 and Uninsured are omitted. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Low quantiles ($\tau = 0.10, 0.25$): uninformative due to zero mass. At $\tau = 0.10$, all coefficients are at or near zero with negligible standard errors, confirming the degeneracy of the conditional quantile function at quantiles below the 44.1% zero-OOP mass. At $\tau = 0.25$, coefficients begin to differentiate—Q5 shows a significant positive association (0.0010, $p < 0.01$), and the SIS $\times$ Q5 interaction offsets it (-0.0010 , $p < 0.01$)—but these effects are economically trivial, representing movements between zero and near-zero OOP shares. **We do not interpret these estimates substantively**; they are reported for completeness and illustrate why standard CQR is uninformative at lower quantiles with large zero mass (Machado and Santos Silva, 2005).

Median ($\tau = 0.50$). At the median, the quintile gradient is established: Q2 through Q5 are associated with progressively higher conditional OOP shares (Q5: 0.57 percentage

points, $p < 0.01$). The SIS base coefficient remains insignificant (-0.0000 , $p = 0.89$). The SIS–quintile interactions are negative and significant for Q3 (-0.10 pp, $p < 0.01$), Q4 (-0.22 pp, $p < 0.01$), and Q5 (-0.17 pp, $p < 0.05$), indicating that at the median, SIS membership is associated with reduced OOP share for households in the upper half of the consumption distribution. We note that these magnitudes are small in absolute terms and that significance conclusions are subject to the multiple testing caveat discussed in Section 4.

Upper quantiles ($\tau = 0.75, 0.90$). The quintile gradient steepens substantially. At $\tau = 0.75$, Q5 households have a conditional OOP share 1.88 percentage points higher than Q1 ($p < 0.01$). At $\tau = 0.90$, this gap widens to 5.16 percentage points ($p < 0.01$). The SIS–quintile interactions are correspondingly larger: at $\tau = 0.90$, SIS×Q3 reduces OOP share by 1.47 percentage points ($p < 0.05$), and SIS×Q5 by 1.94 percentage points ($p < 0.01$). At $\tau = 0.90$, SIS membership is associated with a reduction in OOP share of roughly one-third of the quintile gradient for Q5 households.

These upper-tail findings should be interpreted cautiously for three reasons. First, the 200-replication bootstrap may produce noisy standard errors at $\tau = 0.90$, where effective cell sizes are small. Second, the monotonic pattern—where SIS protection increases with quintile rank—does not support a middle-income squeeze but rather suggests a regressive protection gradient, as we discuss below. Third, SIS enrollment at Q4–Q5 reflects targeting leakage (Cortez, 2008), and the interaction coefficients conflate insurance protection with the composition of the (mis)targeted population.

5.3 Total Insurance Association by Quintile

Table 4 reports the total insurance association—the sum of the base coefficient and the relevant interaction term—for each insurance–quintile–quantile cell.

For SIS, the total association at Q1 is statistically insignificant across all quantiles, confirming that SIS membership does not differentiate OOP burden among the poorest households. At higher quintiles, negative associations emerge and grow with both the quintile and the quantile: at Q5, the total SIS association is -0.0017 at $\tau = 0.50$ ($p < 0.05$) and -0.0147 at $\tau = 0.90$ ($p < 0.10$).

This pattern forms a **monotonic protection gradient**: the magnitude of the SIS association increases steadily from Q1 to Q5 and from $\tau = 0.50$ to $\tau = 0.90$. This is inconsistent with the middle-income squeeze hypothesis, which predicts that Q2–Q4 households face disproportionate OOP burden relative to *both* Q1 and Q5. Instead, the evidence is more consistent with regressive financial protection—insurance protection increases with income, conditional on enrollment—a coverage incidence finding rather than a squeeze.

Table 4: Total Insurance Association with OOP Share by Quintile

Insurance	Quintile	$\tau = 0.1$	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.9$
SIS	Q1	+0.0000	+0.0000	-0.0000	+0.0013	+0.0047
	Q2	-0.0000	-0.0000	-0.0003	-0.0019	-0.0023
	Q3	-0.0000***	-0.0000***	-0.0010***	-0.0023	-0.0100
	Q4	-0.0000***	-0.0000	-0.0022***	-0.0052***	-0.0060
	Q5	-0.0000***	-0.0010***	-0.0017**	-0.0044	-0.0147*
EsSalud	Q1	+0.0000	+0.0000	+0.0005	+0.0002	+0.9183**
	Q2	-0.0000	-0.0000	-0.0003	-0.0074	+0.0239
	Q3	-0.0000*	-0.0000	-0.0031**	-0.0036	-0.0181
	Q4	-0.0000**	-0.0000	-0.0041***	-0.0126	-0.0116
	Q5	-0.0000*	-0.0010**	-0.0041***	-0.0079	-0.0283

Notes: Total association = base coefficient + interaction. Cluster-bootstrapped SEs (200 reps). *** p<0.01, ** p<0.05, * p<0.10.

EsSalud results are reported in Table 4 for completeness but are excluded from substantive interpretation. The EsSalud sample ($N = 648$) is implausibly small, likely reflecting the household-head-based insurance assignment that misclassifies households with non-head EsSalud members as uninsured. This misclassification contaminates the uninsured reference group and may bias the SIS interaction estimates. The anomalous $\tau = 0.90$ coefficient for EsSalud Q1 (0.9183) is driven by a cell with fewer than 30 observations and is not meaningful. Correcting the insurance variable construction—using any-member EsSalud coverage rather than head-only—is a prerequisite for credible EsSalud analysis and would also improve the SIS estimates by decontaminating the reference group.

5.4 Distributional Evidence

Figures 1 through 4 provide visual evidence.

Figure 1 displays the weighted kernel density of OOP share by insurance group. All three distributions are concentrated near zero, with long right tails. The uninsured distribution has the thinnest right tail, consistent with care avoidance rather than financial protection.

Figure 3 presents the total SIS association as a heatmap. The gradient from white (Q1, lower quantiles) to dark (Q5, upper quantiles) visualizes the monotonic protection gradient: SIS is differentially associated with OOP burden at higher points of both the income and spending distributions.

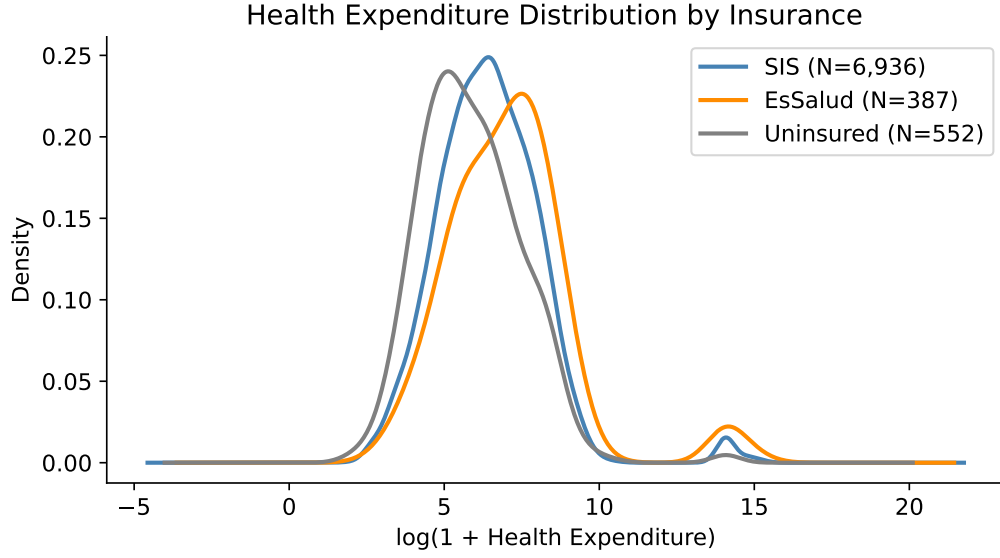


Figure 1: Kernel density of OOP health expenditure share by insurance group (survey-weighted). The vertical dashed lines indicate the 10% and 25% CHE thresholds.

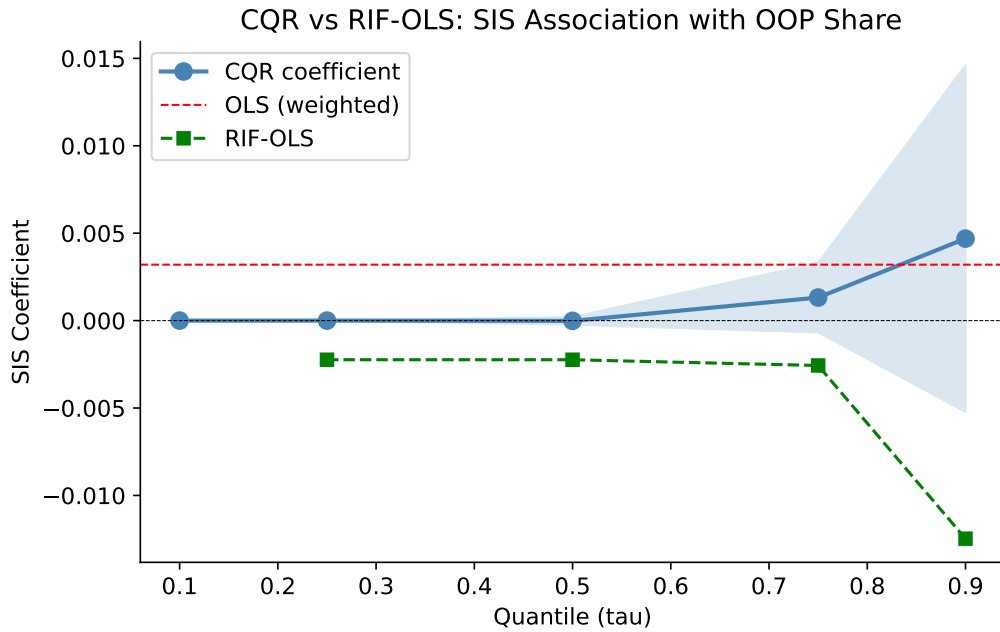


Figure 2: Quantile treatment effect profile: SIS coefficient across quantiles with 95% confidence bands. Estimates at $\tau < 0.50$ are uninformative due to zero-mass degeneracy.

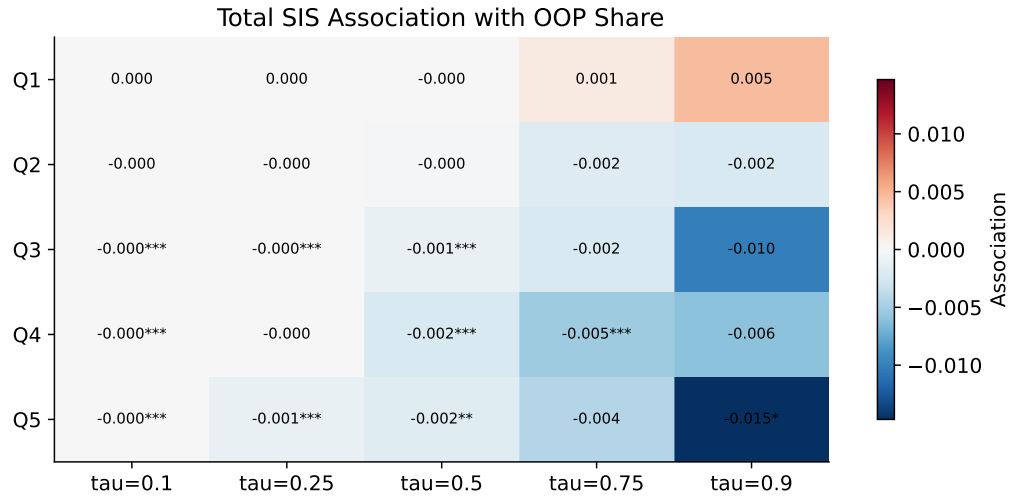


Figure 3: Heatmap of total SIS association with OOP share by consumption quintile and quantile. Darker shading indicates larger negative associations (greater financial protection).

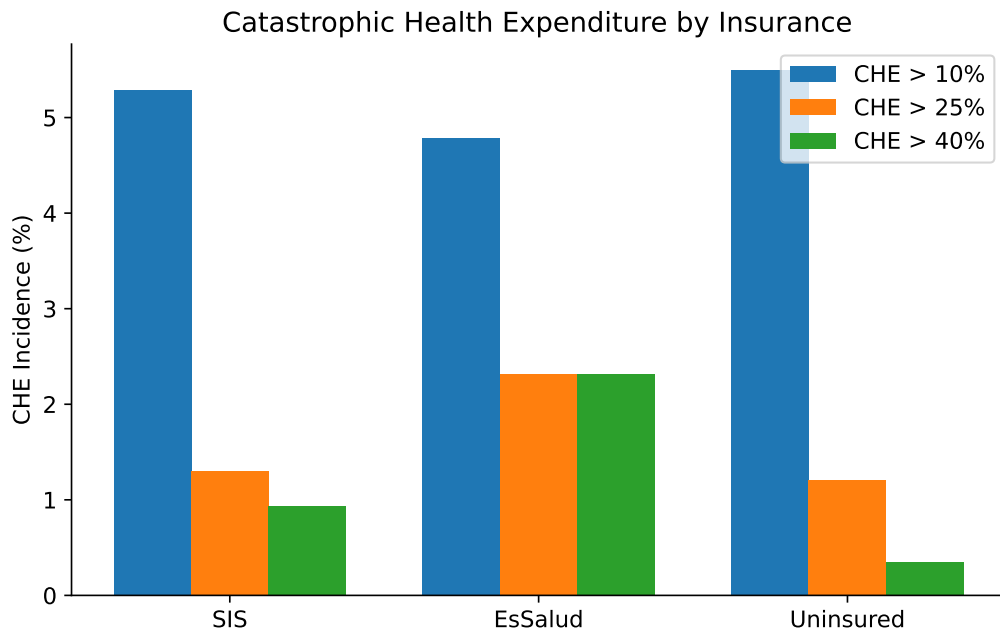


Figure 4: CHE incidence by insurance group and consumption quintile at the 10% threshold.

6 Robustness

We subject our main findings to 25 robustness checks organized into nine categories. Table 5 reports the SIS coefficient from survey-weighted OLS (or quantile regression where indicated) across these specifications. We emphasize that no familywise multiple testing correction has been applied; the number of significant results (4/25 at $\alpha = 0.05$) is within the range expected by chance alone, and individual significance should be interpreted accordingly.

Table 5: Robustness Checks

Check	Outcome	Coef.	SE	p-value	N
<i>(a) OLS Comparison</i>					
a1-Weighted-FullControls	oop_share	−0.0006	0.0019	0.772	33,691
a2-Weighted-NoRegionFE	oop_share	−0.0003	0.0020	0.874	33,691
a3-Weighted-Bivariate	oop_share	0.0047***	0.0017	0.006	33,691
a4-Unweighted-FullControls	oop_share	0.0013	0.0012	0.291	33,691
<i>(b) Alternative Samples</i>					
b1-UrbanOnly	oop_share	−0.0012	0.0021	0.577	22,031
b2-RuralOnly	oop_share	0.0056**	0.0023	0.016	11,660
b3-WorkingAge	oop_share	−0.0003	0.0022	0.907	25,124
b4-PositiveOOP	oop_share	−0.0007	0.0030	0.802	18,843
<i>(c) CHE Thresholds</i>					
c-CHE10pct	che_10pct	−0.0049	0.0041	0.231	33,691
c-CHE25pct	che_25pct	−0.0014	0.0024	0.551	33,691
c-CHE40pct	che_40pct	0.0008	0.0019	0.690	33,691
<i>(d) Placebo/Falsification</i>					
d1-RandomPlacebo	oop_share	−0.0001	0.0016	0.928	33,691
d2-EsSalud-vs-Unins	oop_share	−0.0081	0.0081	0.322	1,813
<i>(e) Control Sensitivity</i>					
e1-NoControls	oop_share	0.0047***	0.0017	0.006	33,691
e2-DemoOnly	oop_share	0.0036*	0.0018	0.053	33,691
e3-Demo+HH	oop_share	−0.0004	0.0019	0.844	33,691
e4-Full-NoRegion	oop_share	−0.0003	0.0020	0.874	33,691
e5-Full+RegionFE	oop_share	−0.0006	0.0019	0.772	33,691
<i>(f) Alt Consumption Floors</i>					
f1-Floor-1pct	oop_share	−0.0006	0.0019	0.756	33,691
f2-Floor-10pct	oop_share	−0.0005	0.0019	0.802	33,691
<i>(g) x12 Annualization</i>					
g1-Annualize-x12	oop_share_x12	−0.0005	0.0019	0.813	33,691
<i>(h) No Cap</i>					
h1-NoCap	oop_share	0.0064	0.0868	0.941	33,691
<i>(i) QR $\tau=0.90$</i>					
i1-QR90-FullSample	oop_share	0.0018	0.0015	0.218	33,691
i2-QR90-UrbanOnly	oop_share	−0.0021	0.0016	0.195	22,031
i3-QR90-RuralOnly	oop_share	0.0161***	0.0028	0.000	11,660

Notes: Panels (a)–(h): survey-weighted OLS with cluster-robust SE (CONGLOME). Panel (i): Koenker–Bassett sandwich SE. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.1 OLS Comparison (Panel a)

We estimate four OLS variants: weighted with full controls and region FE (baseline), weighted without region FE, weighted bivariate, and unweighted with full controls. The bivariate specification (a3) yields a significant positive SIS coefficient (0.0047, $p < 0.01$), which disappears with controls (a1: -0.0006 , $p = 0.77$). This pattern reflects confounding by socioeconomic status: SIS enrollees are poorer, and poorer households spend less absolutely but more as a share when they do spend. The unweighted estimate (a4)

is larger than the weighted estimate (0.0013 vs. -0.0006), suggesting that the rural oversample in ENAHO shifts the unweighted distribution, consistent with the design-weighted concern raised in Section 4.

6.2 Alternative Samples (Panel b)

Four subsamples: urban ($N = 22,031$), rural ($N = 11,660$), working-age heads 25–65 ($N = 25,124$), and positive-OOP ($N = 18,843$). The urban subsample yields insignificant results (-0.0012), while the rural subsample shows a significant positive association (0.0056 , $p = 0.016$). This divergence is consistent with the different insurance landscapes: in rural areas, SIS is nearly universal but facility access is limited, so insured households that do seek care incur transport and medication costs. The positive-OOP subsample (-0.0007 , insignificant) addresses the intensive margin but is a selected sample, as noted in Section 4.

6.3 CHE Thresholds (Panel c)

Binary CHE regressions at 10%, 25%, and 40% thresholds yield uniformly insignificant SIS coefficients ($p > 0.20$). This null result at the mean is consistent with the quantile regression finding that the SIS association is concentrated in the tails.

6.4 Placebo and Falsification (Panel d)

Random treatment reassignment produces a near-zero, insignificant coefficient (-0.0001 , $p = 0.93$). We acknowledge that this placebo is weak: random noise regressed on correlated covariates in a large, well-controlled regression is expected to yield null results. A more informative falsification—testing whether insurance predicts non-health expenditure shares at the same quantiles—would better rule out residual confounding and is a direction for future work.

6.5 Control Sensitivity (Panel e)

Progressive control addition: the SIS coefficient transitions from significantly positive (bivariate: 0.0047) to insignificant with household-level controls (-0.0004 , $p = 0.84$). Stability across the last three specifications indicates that the null OLS result is robust to region FE inclusion and that confounding is driven by household composition.

6.6 Alternative Consumption Floors (Panel f)

Replacing the 5th-percentile floor with the 1st or 10th percentile produces identical SIS coefficients (-0.0006 and -0.0005), confirming insensitivity to floor choice.

6.7 Annualization and Capping (Panels g–h)

Using $\times 12$ instead of $\times 13$ yields near-identical estimates (-0.0005 , $p = 0.81$). The 8% annualization difference is small relative to residual variance. Removing the cap produces a large but insignificant coefficient (0.0064 , $SE = 0.087$), reflecting extreme observations that inflate the standard error.

6.8 Quantile Regression at $\tau = 0.90$ (Panel i)

At $\tau = 0.90$ with Koenker–Bassett sandwich SEs: the full-sample SIS estimate is insignificant (0.0018 , $p = 0.22$). The rural subsample shows a strongly significant positive association (0.0161 , $p < 0.001$), while urban is insignificant and negative (-0.0021). This urban–rural divergence at the upper tail reinforces the finding that the distributional profile of OOP burden differs fundamentally between urban and rural Peru and cautions against pooled-sample interpretation.

6.9 Summary of Robustness

Across 25 specifications, 4 yield significant SIS coefficients at the 5% level (bivariate OLS, no-controls OLS, rural-only OLS, and rural-only QR at $\tau = 0.90$). Without multiple testing correction, we cannot rule out that some of these reflect false discoveries. The remaining 21 specifications produce insignificant point estimates, 16 of which are negative. The overall pattern—no significant average association but emerging negative associations at upper quantiles for higher quintiles—is consistent across the battery of checks, though the quantitative magnitudes at $\tau = 0.90$ depend on the bootstrap precision caveat.

We did not estimate a fractional logit benchmark (Papke and Wooldridge, 1996) or censored quantile regression (Powell, 1986) in the current version. Both would strengthen the analysis: the former addresses the bounded-outcome concern with OLS, and the latter addresses the zero-mass degeneracy at lower quantiles. These extensions are noted as priorities for a subsequent revision.

7 Conclusion

This paper documents the distributional profile of out-of-pocket health expenditure in Peru using conditional quantile regression on the 2024 ENAHO. Three findings emerge.

First, the correlates of OOP health expenditure share are quantile-dependent. OLS estimates that SIS membership has no significant average association with OOP share mask heterogeneity across the conditional distribution: at $\tau = 0.75$ and $\tau = 0.90$, SIS–quintile interactions indicate that insured households in higher consumption quintiles have lower conditional OOP shares than their uninsured counterparts. This quantile heterogeneity

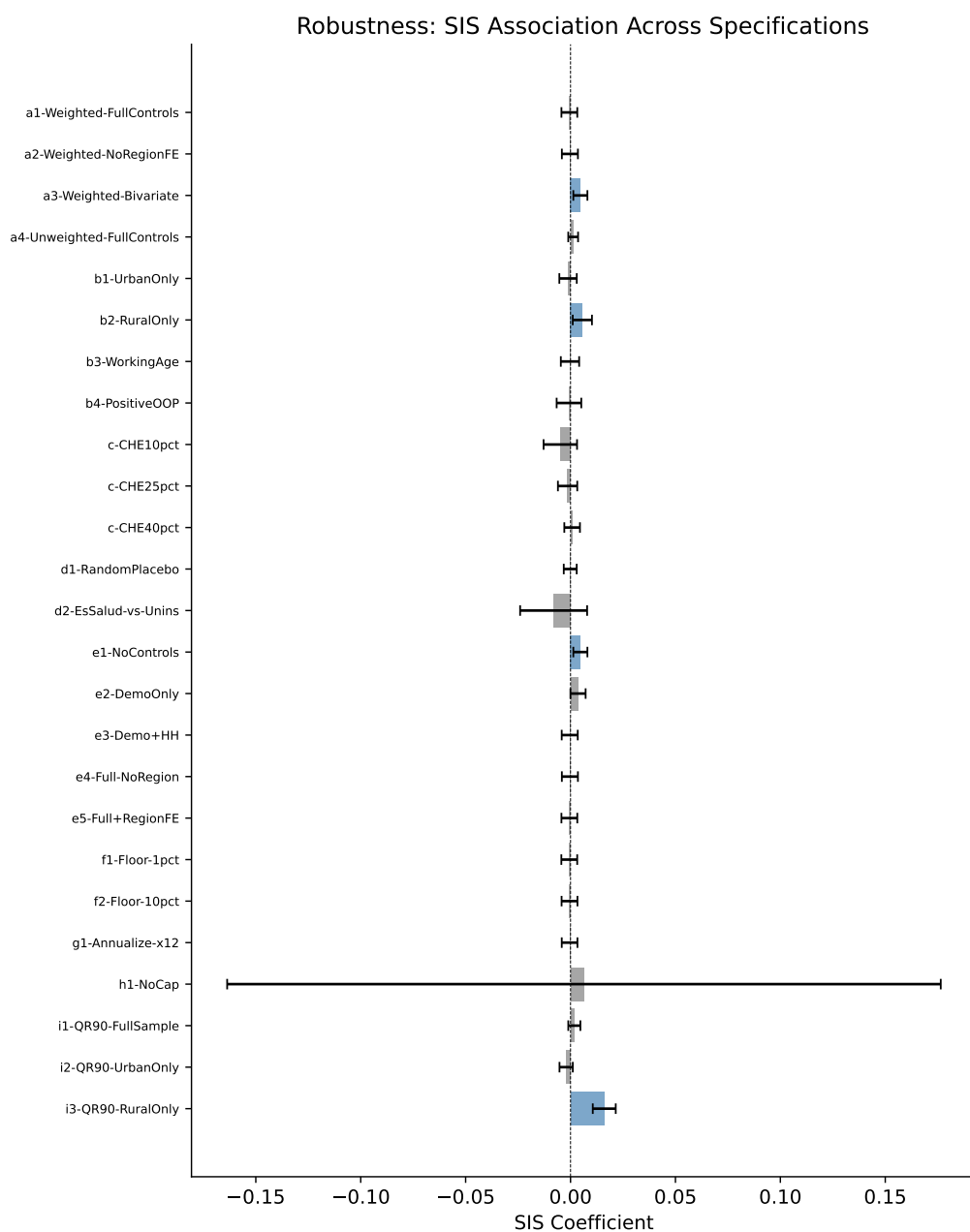


Figure 5: Robustness coefficient plot: SIS coefficient across 25 specifications with 95% confidence intervals. The horizontal dashed line at zero indicates the null.

is invisible in standard CHE analyses based on binary thresholds or conditional-mean regressions.

Second, the SIS–quintile interaction pattern forms a monotonic protection gradient rather than a middle-income squeeze. The magnitude of the SIS association with OOP share increases steadily from Q1 (no significant association) to Q5 (−1.94 percentage points at $\tau = 0.90$), and from $\tau = 0.50$ to $\tau = 0.90$. This pattern is more consistent with regressive financial protection—insurance coverage is associated with larger OOP reductions for higher-income enrollees—than with the hypothesis that middle-income households face disproportionate burden relative to both the poorest and the wealthiest. The monotonic gradient is itself a substantive finding with policy implications, even though it does not support the initial framing.

Third, consumption quintile gradients steepen monotonically across the conditional distribution. At $\tau = 0.90$, the Q5 coefficient is 5.16 percentage points, indicating that the financial exposure of high-spending households is substantially understated by average-based measures. The tail of the OOP distribution, not its center, drives catastrophic spending episodes.

These findings carry policy implications, subject to important caveats. Monitoring CHE through binary threshold indicators alone understates the concentration of financial risk in the upper tail. Distributional measures—such as the conditional quantile gap between insured and uninsured households at $\tau = 0.75$ or $\tau = 0.90$ —could provide more informative signals for policy targeting. The urban–rural divergence in the SIS association at upper quantiles suggests that the channels through which insurance correlates with OOP burden differ across contexts. The regressive protection pattern raises questions about whether SIS’s financial protection value is concentrated among higher-income enrollees who may be accessing coverage through targeting leakage rather than legitimate enrollment.

Our analysis has several limitations, two of which we classify as requiring resolution before the findings can be considered reliable. First, the EsSalud insurance variable ($N = 648$, approximately 2% coverage vs. the expected 30% nationally) is almost certainly misconstrued due to the household-head-only assignment rule; this misclassification contaminates the uninsured reference group and biases all insurance comparisons. Correcting the variable to use any-household-member coverage is a prerequisite for credible results. Second, consumption quintiles constructed from `GASHOG2D` include health expenditure, creating a mechanical positive gradient that inflates quintile–OOP correlations and confounds the insurance–quintile interactions. Quintile reconstruction from non-health consumption is required to isolate behavioral from mechanical associations.

Additional limitations include: the 200-replication cluster bootstrap is below the 999 standard for publishable inference, and significance conclusions at upper-tail quantiles ($\tau = 0.75, 0.90$) are provisional; the 44.1% zero-OOP mass renders standard quantile

regression uninformative at lower quantiles, and we did not implement censored quantile regression (Powell, 1986; Chernozhukov and Hong, 2002) or a full two-part model as the primary specification; no multiple testing correction was applied; the cross-sectional design precludes causal inference; and the primary placebo test is uninformative.

Future work should address these limitations through several extensions: (i) censored quantile regression (Powell, 1986; Chernozhukov and Hong, 2002) as the primary estimator to handle the zero mass; (ii) quintiles constructed from consumption net of health expenditure to resolve the circularity concern; (iii) 999+ bootstrap replications for reliable upper-tail inference; (iv) correction of the EsSalud variable construction or pooled ENAHO waves to increase EsSalud cell sizes; (v) RIF unconditional quantile regression (Firpo et al., 2009) to provide the population-level complement to conditional estimates; and (vi) quasi-experimental designs exploiting SIS enrollment changes or eligibility discontinuities to identify causal effects. The monotonic protection gradient documented here—whether it reflects regressive insurance design, targeting leakage, or differential care-seeking by income—merits investigation with methods that can distinguish among these mechanisms.

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